A Cloud-Native 5G Testbed with an SDN Transport for Anomaly Detection and Handover Prediction

Rihem Fathallah
ENET'Com, University of Sfax, Tunisia
rihem.fathallahengineer@gmail.com

***Abstract*—We build an emulated 5G standalone network (Open5GS core, srsRAN gNB and UE over a ZeroMQ virtual radio) in a single-node Kubernetes cluster, whose N3 user plane crosses a software-defined transport ring. Faults injected into the transport and into the network functions yield a labelled dataset of [TBD: N] hours with [TBD: N] faults. We compare a z-score baseline, Isolation Forest and an LSTM autoencoder at event level (recall, detection delay, false alarms per hour) and apply them to public BGP data. Separately, an LSTM handover predictor is compared with a 3GPP A3 rule tuned per speed on TR 38.901 signal traces. [TBD: headline results].**

I. Introduction

Research questions:

- **RQ1** On a Kubernetes-hosted Open5GS core with an SDN transport, what event recall, false alarms per hour and median detection delay do Isolation Forest and an LSTM autoencoder reach against a z-score baseline, per fault type and severity?
- **RQ1b** Do the same models and features detect known anomalous events in public BGP data?
- **RQ2** Does an LSTM predictor reduce ping-pong and radio-link-failure rates against an A3 rule tuned per speed, without raising the handover rate?

Contributions: [TBD: after results].

II. Testbed

A. Architecture

[TBD: architecture figure]. Open5GS v2.7.7 (10 network functions of the 5G service-based architecture [1]), srsRAN Project gNB and srsUE over ZeroMQ, single-node k3s with the static CPU manager: the gNB, UE and UPF containers get exclusive cores so that CPU faults cannot spill onto the real-time radio emulation or the measurement path.

B. SDN transport

N3 crosses a four-switch Open vSwitch ring (Mininet) controlled by Ryu. The controller computes a breadth-first forwarding tree rooted at the RAN-site switch and recomputes it on every port-status change; N3 normally follows s1–s2–s3 and is rerouted over s4 when a link on that path fails (measured reroute: [TBD: host value] s). A spanning-tree design was rejected: Ryu's stplib did not reconverge after a failure on the active path.

TABLE I
Fault catalogue

Type	Injector	Mild	Strong
delay	Mininet link	10 ms	50 ms
loss	Mininet link	1 %	5 %
link down	Mininet link	–	–
CPU	Chaos Mesh (UPF)	1 worker, 50 %	2 workers, 100 %
memory	Chaos Mesh (UPF)	256 MB	768 MB
pod kill	Chaos Mesh (AMF/SMF)	–	–
flood	UE iperf3	4×, 30 Mb/s	8×, 80 Mb/s

C. Monitoring

Prometheus scrapes, every 5 s: Open5GS NF metrics, kubelet/cAdvisor, node-exporter, the controller's port and tree metrics, and a UE-side probe (RTT, loss, tunnel throughput, registration).

III. Fault model and dataset

Table I: fault types, injection point (one per type), mild/strong parameters. Ground truth uses the times at which an impairment was *confirmed* active and removed; the effect window adds a 60 s recovery grace. Windows (60 s, stride 5 s) are anomalous if $\geq 50\%$ of their samples fall in an effect window, normal if none, ambiguous otherwise (excluded from window metrics). Splits are by time per run (60/20/20) with a 60 s purge at boundaries. [TBD: runs, hours, faults per type and severity, failed injections].

IV. Anomaly detection (RQ1, RQ1b)

Models: z-score ($|z| > 3$), Isolation Forest [2] (threshold: best validation F1), LSTM autoencoder (threshold: 99th percentile of validation scores on normal windows) [3], Random Forest as a supervised upper reference. Metrics at event level (recall, delay from t_{applied} to the end of the first alarm window), false alarms per hour of normal operation (alarms merged within 30 s), and window-level F1. No point adjustment [4]. BGP features follow [5]. 5 seeds, mean ± std. [TBD: Fig. detection_summary, detection_by_type, detection_delay, attribution; ablation without link-state columns; RAN transfer; BGP table].

V. Handover prediction (RQ2)

A. Simulation model

Seven omnidirectional cells (ISD 500 m), TR 38.901 [6] UMa path loss (verified identical to ns-3's implementation),

spatially correlated shadowing, a smooth LOS field with soft transitions, UEs at 3/30/60/120 km/h, 40 ms RSRP samples with 1 dB measurement error, TS 38.331 [7] L3 filter ($k = 4$, coefficient rescaled to keep the 200 ms time constant). Full-load SINR; RLF via $Q_{\text{out}} = -8$ dB, $Q_{\text{in}} = -6$ dB, T310 = 1 s; ping-pong = return within 1 s.

B. Policies and fairness

A3 is tuned per speed over hysteresis 0–10 dB and TTT $\{40,160,320,640\}$ ms; the LSTM's threshold and minimum time-of-stay are tuned per speed with the same objective J = RLF/min + 0.5 PP/min + 0.05 HO/min on the same validation trajectories. Label: a neighbour exceeds the serving cell by ≥ 3 dB for ≥ 200 ms starting within 500 ms (noise-free large-scale RSRP). Training states come from A3 rollouts plus one DAgger round. [TBD: Fig. handover_rates, a3_grid; paired bootstrap CIs (LSTM − A3) per speed].

VI. Limitations

Emulated radio with a single UE (srsRAN) or protocol-level UEs (UERANSIM); synthetic faults; a single host; simplified mobility (omnidirectional cells, no fast fading); BGP intervals taken from the literature. Results describe the core and transport behaviour of an emulated network.

References

[1] "System architecture for the 5G system," 3GPP, Tech. Rep. TS 23.501.

[2] F. T. Liu, K. M. Ting, and Z.-H. Zhou, "Isolation forest," in *Proc. IEEE ICDM*, 2008.

[3] P. Malhotra *et al.*, "LSTM-based encoder-decoder for multi-sensor anomaly detection," in *ICML Anomaly Detection Workshop*, 2016.

[4] S. Kim, K. Choi, H.-S. Choi, B. Lee, and S. Yoon, "Towards a rigorous evaluation of time-series anomaly detection," in *Proc. AAAI*, 2022.

[5] N. Al-Rousan and L. Trajković, "Machine learning models for classification of BGP anomalies," in *Proc. IEEE HPSR*, 2012.

[6] "Study on channel model for frequencies from 0.5 to 100 GHz," 3GPP, Tech. Rep. TR 38.901.

[7] "NR; radio resource control (RRC) protocol specification," 3GPP, Tech. Rep. TS 38.331.